

Selma Baraka

PhD Student in Electrical Engineering

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Profile

PhD Student in Electrical Engineering at Budapest University of Technology and Economics (BME) under the supervision of Prof. David Raisz. My research focuses on physics-based and data-driven digital twins for electrochemical energy storage systems, with particular emphasis on lithium-ion battery degradation modelling, state-of-health estimation, prognostics, and machine learning applications for energy systems. I have consistently ranked first throughout my undergraduate and master's studies and am passionate about developing intelligent, reliable, and explainable solutions for next-generation energy storage technologies.

Research Interests

- Battery Digital Twins
- Electrochemical Energy Storage Systems
- Lithium-Ion Batteries
- Battery Modelling
- Battery Degradation Analysis
- State-of-Health (SOH) Estimation
- Machine Learning and Deep Learning
- Physics-Informed Neural Networks (PINNs)
- Smart Grids
- Renewable Energy Systems

Education

Budapest University of Technology and Economics (BME), Hungary

PhD in Electrical Engineering

Sep 2026 – Present

Research Supervisor: Prof. David Raisz

Research Topic: Physics-Based and Data-Driven Digital Twin for Electrochemical Energy Storage Systems

University Djillali Liabès, Algeria

Master of Science (MSc) in Electrotechnics (MCIL)

Oct 2024 – Jun 2026

- Rank: 1st / 25

- GPA: 16.26 / 20
- Thesis: Design and Implementation of an Intelligent Automatic Waste Sorting System Based on AI

University Djillali Liabès, Algeria

Bachelor of Science (BSc) in Electrotechnics

Oct 2021 – Jun 2024

- Rank: 1st / 33
- GPA: 17.08 / 20
- Thesis: Electromagnetic Compatibility Compliance Test

Tahiri Moubarak High School

Baccalaureate in Experimental Sciences

Jul 2021

- GPA: 16.02 / 20
- Mention: Very Good

Current Research

My current PhD research aims to develop a physics-based and data-driven digital twin framework for lithium-ion batteries. The work combines electrochemical modelling, machine learning, and battery health diagnostics to improve state-of-health estimation, degradation prediction, and predictive maintenance of energy storage systems.

Professional Experience and Internships

SETRAM SPA

Research Intern

Jan 2026 – Feb 2026

- Analyzed energy efficiency parameters for electrotechnical systems in urban transit.

NAFTAL

Engineering Intern

17 May 2026 – 31 May 2026

- Observed industrial energy distribution and fuel storage operations.
- Participated in technical monitoring and maintenance activities.
- Acquired practical knowledge of industrial safety procedures and HSE requirements.

ENIE (National Enterprise of Electronic Industries)

Intern

Mar 2024 – Apr 2024

- Gained hands-on experience in the Electronic Integration Unit.
- Assisted in hardware testing and quality control processes.

Sonelgaz Distribution

Intern

Dec 2023 – Jan 2024

- Explored power distribution network operations.
- Observed maintenance protocols and grid stability management.

Research Projects

Physics-Based and Data-Driven Digital Twin for Electrochemical Energy Storage Systems

- Development of advanced battery digital twin methodologies.
- Machine learning approaches for battery degradation modelling.
- SOH estimation and predictive prognostics.
- Data-driven diagnosis of energy storage systems.

AI-Based Waste Sorting System

- TensorFlow Lite and Raspberry Pi implementation.
- Computer vision-based material classification.
- Industrial automation integration.

Technical Skills

Electrical Engineering

- Power Systems
- Energy Storage Systems
- Smart Grids
- Renewable Energy Systems
- Power Electronics
- Control Systems

Artificial Intelligence

- Machine Learning
- Deep Learning
- Computer Vision
- Physics-Informed Neural Networks (PINNs)
- Data Analysis

Software and Programming

- MATLAB / Simulink
- Python
- TensorFlow
- PowerWorld Simulator

- PSIM
- SolidWorks
- Arduino
- Raspberry Pi
- Git
- GitHub

Languages

- Arabic: Native
- French: Full Professional Proficiency
- English: Advanced Professional Proficiency
- IELTS Academic: Overall Band Score 6.0

Awards and Certifications

- 4 Stars Excellence Recommendation (2026)
- Top Ranking Attestation (1st Place)
- HSE Certification (2025)
- Career Essentials in Generative AI – Microsoft and LinkedIn Learning (June 2026)
- Getting Started with Deep Learning – NVIDIA (June 2026)
- Workshop Telco by Djezzy and Huawei (2025)

Extracurricular Activities

- Member of the Geniuses Club.
- Active participation in scientific outreach and innovation activities.